

# ADEBAYO A. OMOLUMO

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## SUMMARY

Results-driven Senior DevOps & Cloud Engineer (AWS Specialist) with 8+ years of experience architecting, automating, and scaling secure enterprise-grade cloud infrastructure. Expert in AWS architecture, Infrastructure as Code (Terraform, CloudFormation), CI/CD, Kubernetes (EKS), and automation with Python/Bash. Proven record of leading cloud migrations, optimizing multi-tier architectures, enforcing security compliance, and mentoring engineering teams to adopt DevOps best practices. Holds multiple cloud certifications (AWS, Azure, GCP) and a PMP, demonstrating both technical depth and project leadership. Passionate about building resilient systems, cost-efficient AWS solutions, and enabling developer velocity through automation and observability.

## SKILLS

- **Cloud Platforms:**
  - **AWS (Primary)** — EC2, EKS, ECS, Lambda, S3, RDS, CloudFront, IAM, VPC, Route 53, CloudFormation, CloudWatch, KMS, Secrets Manager, Systems Manager, CloudTrail
  - Azure – AKS, App Service, Azure DevOps, Sentinel, Logic Apps, Event Hubs, Functions
  - GCP – GKE, Cloud Run, Vertex AI, Cloud Storage, Cloud Functions, BigQuery, Vertex AI, Cloud Run
- **DevOps & CI/CD:** Jenkins, CircleCI, Azure DevOps, GitHub Actions, Docker, Kubernetes (EKS, Kubespray), Ansible, Terraform, CloudFormation, Prometheus, Grafana, ELK Stack, Git
- **Programming & Scripting:** Python (Flask, Django), JavaScript (React.js, Next.js, Vue.js, Node.js), TypeScript, Solidity, Bash, PowerShell, HTML5, CSS3
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Cosmos DB, SQL Server
- **Operating Systems & Networking:** Linux (Ubuntu, CentOS), Windows Server, TCP/IP, DNS, VPN, Load Balancing, Firewalls
- **Security & Compliance:** IAM, Network Security, Data Encryption, SIEM (Azure Sentinel), Security Best Practices, PCI-DSS Awareness, Vulnerability Management, Zero Trust Principles
- **Tools & Methodologies:** Agile, Scrum, Jira, Confluence, Microsoft Office Suite, Project Management

## PROFESSIONAL EXPERIENCE

### Senior Cloud DevOps Engineer | INFOMETICS

September 2024 – Present

- **Architected and Deployed a Production-Grade, Highly Available Kubernetes Cluster on AWS:**

- Designed and built a high-fidelity simulation of a secure, air-gapped, on-premise environment to validate **stateful IBM workloads (ACE/MQ)**.
- Evolved the entire cluster lifecycle management from initial shell-script-based bootstrapping to a fully automated, idempotent deployment using **Ansible** and **Kubespray**, demonstrating a strategic progression towards enterprise-grade automation.
- **Infrastructure as Code (IaC) & Automation:**
  - **Initial State:** Provisioned a multi-node Kubernetes cluster using Terraform, with nodes configured via user-data shell scripts and **kubeadm**.
  - **Final State:** Architected a fully automated, production-grade deployment using **Kubespray (Ansible)**, migrating all OS-level and Kubernetes bootstrapping logic from fragile scripts to a robust, declarative, and community-vetted framework. Provisioned a **6-node** (3 master, 3 worker) HA cluster.
- **High-Availability Control Plane Engineering:**
  - **Initial State:** Managed a single-master control plane.
  - **Final State:** Engineered a fully resilient, multi-master (3-node) control plane using **kube-vip** for automated API server load balancing. Systematically diagnosed and resolved complex AWS networking issues, including VPC routing and security group rules, to enable cross-subnet VIP reachability.
- **Distributed Storage & Stateful Workloads:**
  - Implemented a resilient, software-defined storage layer using **Longhorn** for all stateful applications.
  - Orchestrated IBM ACE and MQ applications as robust StatefulSets with persistent, replicated storage, ensuring data durability and stable network identity.
- **Secure & Observable Networking:**
  - Engineered a comprehensive ingress and pod-to-pod communication strategy using **NGINX Ingress** and **Calico CNI**.
  - Implemented a **zero-trust** network model by authoring granular **NetworkPolicies** and correctly configuring host-level firewalls (**firewalld**) to secure all application traffic while ensuring full cluster functionality.
- **Disaster Recovery & Full-Stack Observability:**
  - Designed and validated a complete disaster recovery solution using **Velero** and **Minio**.

- Deployed and configured a comprehensive monitoring stack with **Prometheus** and **Grafana**.
- Lead the setup and management of testing and development environments, ensuring scalable cloud infrastructure, seamless workflows, and optimized resource allocation with security best practices.
- Develop automated backup systems for code repositories (e.g., using AWS Lambda and Docker), ensuring weekly backups with robust logging and security compliance.
- Migrate frontend, backend, and database Docker applications to secure networks, maintaining compliance with relevant security policies.
- Implement automation on cloud platforms (e.g., AWS) to manage testing databases and instances, significantly reducing resource usage and associated costs.
- **Key Client Engagement: UBA Group (United Bank for Africa), Lagos**
  - Engineered and deployed containerized (Docker) test environments for Nginx, frontend, and backend applications within UBA's secure, isolated internal network.
  - Configured and managed server installations with all necessary dependencies, enabling multiple distinct projects to run concurrently on single test servers via containerization, each on dedicated ports.
  - Spearheaded the seamless migration of applications from the meticulously prepared test environment to the live production environment.
  - Led the critical upgrade of the project's network infrastructure to HTTPS, significantly enhancing data security and integrity.
  - Designed and implemented a custom deployment pipeline to build, package, and securely transfer frontend and backend artifacts to the air-gapped bank environment, overcoming internet access restrictions.
  - Collaborated extensively with UBA's security team to implement and enforce stringent security best practices, including HSTS, SSL certificate management, secure http-only cookies, and comprehensive Nginx security policies.

**Senior Cloud/DevOps Engineer** | IGLUM TECH SERVICES (formerly Iglum Nig. Ltd.) | Ikorodu, Lagos  
*September 2019 – August 2024*

- Founded and led a tech services agency, evolving from full-stack web development to specialized cloud engineering and DevOps consulting for various clients.
- Led the design and deployment of cloud infrastructure on AWS, Azure, and GCP, ensuring scalability, security, and compliance with industry standards for client projects.

- Architected and implemented CI/CD pipelines using Jenkins, CircleCI, and Azure DevOps for automated build, test, and deployment of web applications and microservices, reducing deployment time by up to 50%.
- Managed migration of on-premise infrastructure to Azure for clients, optimizing resource allocation and achieving average annual cloud expense reductions of 30%.
- Developed and launched "Save Earth," an AI-powered platform for endangered species awareness, integrating Azure Custom Vision and Blob Storage; secured two hackathon wins and AWS funding.
- Redesigned and migrated the Iglumtech company website to AWS S3 with CloudFront CDN, implementing a CircleCI DevOps workflow for automated updates and enhanced security (HTTPS).

**Computer Science Teacher** | National Youth Service Corps – UEC College | Ikorodu, Lagos  
*September 2019 – March 2020*

- Taught computer science fundamentals and programming concepts to secondary school students.

## KEY PERSONAL PROJECTS

- [DockerHelper](#) (**AI-powered Docker config generator**): Developed a Python FastAPI backend tool leveraging Large Language Models to automatically generate optimized Dockerfiles and docker-compose configs by analyzing Git repos. Implemented OAuth authentication, a Retrieval-Augmented Generation pipeline for continuous learning, and multi-provider OAuth integration. Future plans include frontend enhancements and CI/CD integration. *(September 2025)*
- [SmartWait](#) (**Virtual queue management system for healthcare**): Built a full-stack web app enabling patients to check in remotely, get real-time SMS queue updates, and help staff manage patient flow via a secure dashboard. Used Node.js, Prisma ORM, and Socket.io for real-time communication. Collaborated with an AI partner for architecture and code generation. Planned expansions include analytics and ML-based wait time predictions. *(May 2025)*
- **Chainlink Oracle Integration (Hackathon Winner)**: Led a team to streamline Chainlink project deployments with CircleCI, AWS CloudFormation, automated rollback, React/Next.js frontend, and Solidity/Hardhat backend. *(May 2022 - Aug 2022)*
- **Operationalizing ML Microservice with Kubernetes**: Deployed a Scikit-learn model as a microservice using Docker, Kubernetes, and CircleCI, improving logging and testing. *(Sept 2023 - Oct 2023)*

- **Azure Cloud Security & Monitoring Implementation:** Led client projects enhancing security with Azure Sentinel, Security Center, Zero Trust networking, and automated remediation via Azure Runbooks. *(Aug 2021 - Sept 2022)*

## **CERTIFICATIONS & PROFESSIONAL DEVELOPMENT** (Selected)

- **Project Management Professional (PMP®)** – PMI
- **Microsoft Certified: DevOps Engineer Expert** (Earned Aug 2023, Renewed)
- **Google Cloud Certified: Professional Cloud Security Engineer** (Earned June 2024)
- **Google Cloud Certified: Professional Cloud Architect** (Earned Nov 2022)
- **Google Cloud Certified: Associate Cloud Engineer** (Earned Apr 2022, Renewed)
- **Microsoft Certified: Azure Administrator Associate** (Earned Apr 2021, Renewed)
- **Microsoft Certified: Azure Developer Associate** (Earned Aug 2021, Renewed)
- **Microsoft Certified: Azure Fundamentals** (Earned Apr 2021)
- **Udacity Nanodegree: Cloud Developer using Microsoft Azure** (Sept 2021)
- **LFS211: Linux Networking and Administration** – The Linux Foundation (Mar 2024)
- **AWS Educate Machine Learning Foundations** (Valid through Apr 2025)
- **Multiple Google Cloud Skill Badges** (*Vertex AI, Security, Data Loss Prevention, etc.* – 2024)
- **OpenHack: DevOps** – Microsoft (May 2021)

## **EDUCATION**

### **Bachelor of Engineering (B.Eng.), Wood Products Engineering**

University of Ibadan | Ibadan, Oyo State, Nigeria

2015 – 2019

### **National Diploma, Mechanical Engineering**

Lagos State Polytechnic | Ikorodu, Lagos State, Nigeria

2013 – 2015

## **AWARDS & RECOGNITION** (Selected)

- **Bunnysell Hackathon Winner** (Team Lead) – 2023
- **Celo's Make Crypto Mobile Hackathon Winner** (Infrastructure & Web3 Track, 1st Place) – 2021
- **Datastax Holiday Hackathon Winner** (EMEA Category) & Top 10 Finalist (4th) – 2021

- **SaveEarth Project: AWS Funding Recipient & 2x Hackathon Winner** – 2020-2022
- **Microsoft Azure AI Hackathon Top 50 Finalist (23rd)** – 2021
- **Andela Google Africa Developers Scholarship (GCP Track Mentor)** – 2020-2021
- **Bertelsmann Technology Scholarship Program (Azure Cloud Computing)** – 2020-2021